

WASP-11b

R-2133

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_101122 · created 2026-08-11T23:38:02+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2455522.795514 +/- 1.5e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.05
Transit depth (Rp/Rs)^2	3.1 +/- 1.77 [%]
Orbital Inclination (inc)	89.36 +/- 1.69
Airmass coefficient 1 (a1)	0.31 +/- 0.18
Airmass coefficient 2 (a2)	0.74 +/- 1.29
Scatter in the residuals of the lightcurve fit is	58.46 %
Best Comparison Star	#1 - [595, 321]
Optimal Aperture	1.82
Optimal Annulus	5.75
Transit Duration (day)	0.0745 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.176 $\pm$ 0.05 vs 0.131 (deviation 0.9 σ)
Duration fit vs published	0.0745 d vs 0.1075 d (deviation 5.0 σ)
Mid-time O-C	-0.2 min
Depth SNR	3.5
Residual scatter	58.46 %

Light curve

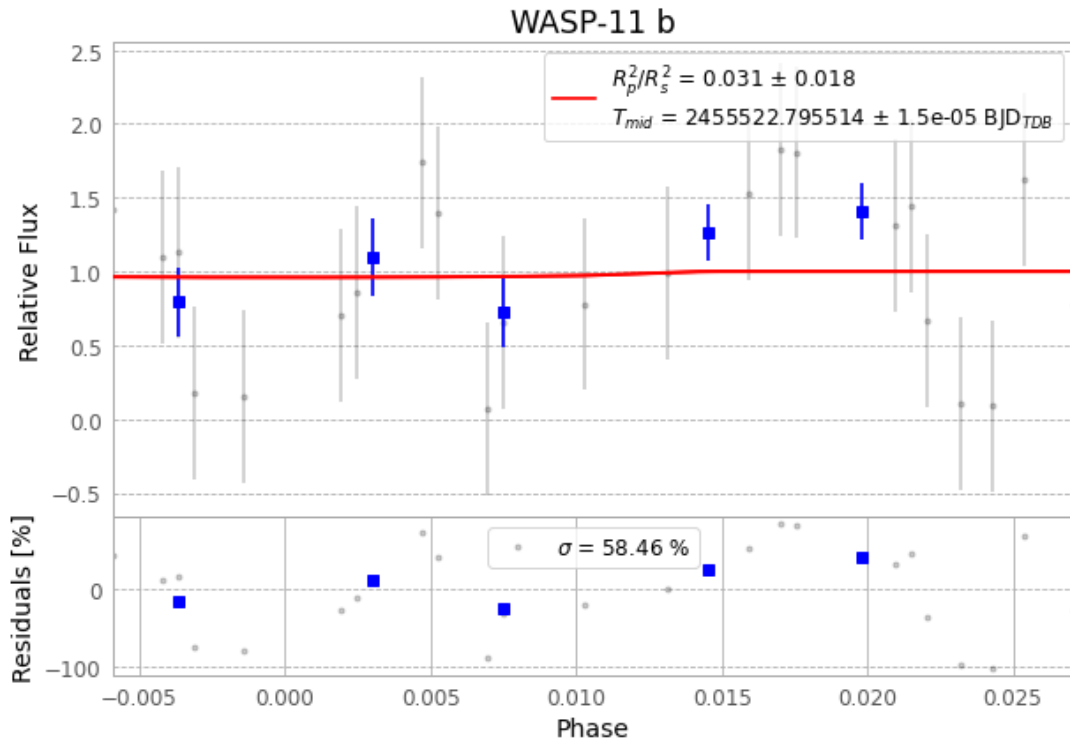


Figure 1 — FinalLightCurve_WASP-11 b_2010-11-21.png (SHA-256 2b516cf0253ffd86...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [412, 437], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 25c0b18d6dd705f4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

66 FITS frames (43 MB), HATP-10101122060614.FITS → HATP-10101122092113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-101122.log (fe866b0d9237...), FinalLightCurve_WASP-11 b_2010-11-21.png (2b516cf0253f...), FinalLightCurve_WASP-11 b_2010-11-21.csv (f83a9d16356f...)

Compute resources

Wall time 142.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 23:38Z.